EE1101

Quiz 2

Disclaimer:

This paper has been built by reverse-engineering the answer paper after evaluation. The questions are only approximate replicas of the original paper.

1. Consider a signal $x(t)$ which consists of only the positive part of a sine wave with a time period of 1 second. Find:

   a) The Fourier series coefficients and Fourier series representation of $x(t)$. **(4 marks)**

   b) The infinite sum $\sum_{k=1}^{\infty} \frac{1}{(4k^2-1)^2}$ **(3 marks)**

   c) The Fourier series coefficients of $x(t) + x(t - \frac{1}{2})$ **(2 marks)**

2. Consider a system with impulse response $h(t) = sin(t)$. Verify if the system is memoryless, stable, causal and invertible.
   (4 marks)

3. Consider a system with $H(j\omega) = \frac{1-j\omega}{1+j\omega}$. Let the output $y(t)$ be obtained when the system is given an input $x(t) = sin(\pi t) + sin(2\pi t)$.
   Find and plot the magnitude and phase of the Fourier coefficients of $y(t)$.
   (7 marks)

1. Consider a signal $x(t)$ which consists of only the positive part of a sine wave with a time period of 1 second. Find:

   a) The Fourier series coefficients and Fourier series representation of $x(t)$. **(4 marks)**

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   c) The Fourier series coefficients of $x(t) + x(t - \frac{1}{2})$ **(2 marks)**

$\int_0^{1/2} \sin 2\pi t\, dt = -\left|\frac{\cos(2\pi t)}{2\pi}\right|_0^{1/2}$ $\quad \frac{+1+1}{2\pi}$

$a_k e^{-jk\pi \cdot 1/2}$ $\quad a_k = (-1)^k a_k$

$x(t) = \sin 2\pi t \quad t \in [0, 1/2]$ $\quad a_0 = 2/\pi$

$0 \quad t \in [1/2, 1]$

$$a_n = \frac{1}{1}\int_0^1 x(t) e^{-jn\omega_0 t} = \int_0^{1/2} \sin 2\pi t\, e^{-jn\cdot 2\pi t}$$

$$= \int_0^{1/2} \frac{e^{j\cdot 2\pi t - j\cdot n\cdot 2\pi t} - e^{-j\cdot 2\pi t - j\cdot n\cdot 2\pi t}}{2j} dt$$

$$= \frac{1}{2j}\int_0^{1/2} e^{2\pi j t[1-n]} - e^{2\pi j t[1-n]}\, dt$$

$$= \frac{1}{2j}\left[\frac{e^{2\pi j t(1-n)}}{2\pi j(1-n)}\Big|_0^{1/2} - \frac{e^{2\pi j t(-1-n)}}{2\pi j(-n-1)}\Big|_0^{1/2}\right]$$

$$= \frac{1}{2j}\left\{\frac{e^{\pi j(1-n)} - 1}{2\pi j(1-n)} + \frac{e^{\pi j(-1-n)} - 1}{2\pi j(1+n)}\right\}$$

$$a_n = \frac{-1}{4\pi}\left\{\frac{e^{\pi j(1-n)} - 1}{1-n} + \frac{e^{\pi j(-1-n)} - 1}{1+n}\right\}$$

$n = \text{odd}: \; a_n = 0 \quad (n \neq \pm 1)$

$n = \text{even}: \; a_n = \frac{-1}{4\pi}\left[\frac{-2}{1-n} - \frac{2}{1+n}\right] = \frac{1}{2\pi}\left[\frac{-2}{n^2-1}\right] = \underline{\underline{\frac{-1}{\pi(n^2-1)}}}$

$n=1$ $\quad a_k = \int_0^{1/2} x(t) e^{-j\cdot 2\pi t}\, dt = \int_0^{1/2} \frac{e^{j\cdot 2\pi t} - e^{-j\cdot 2\pi t}}{2j} \cdot e^{-2\cdot j\pi t}\, dt$

$= \int_0^{1/2} \frac{1 - e^{-4\pi j t}}{2j}\, dt = 1/4j - 1/2j \int_0^{1/2} e^{-4\pi j t}\, dt$ (the integral $\to 0$)

$n=-1 : a_k = -1/4j$

$x(t) = \sum_{\substack{n=-\infty \\ n\neq \pm 1}}^{\infty} \frac{1+(-1)^n}{2\pi(1-n^2)} e^{jn\cdot 2\pi t} + 1/4j\, e^{j\cdot 2\pi t} - 1/4j\, e^{-j\cdot 2\pi t}$

$x(t) = \sum_{k=-\infty}^{\infty} \frac{1}{\pi(1-4k^2)} e^{j4\pi k} + \frac{1}{4j} e^{j\cdot 2\pi t} - 1/4j\, e^{-j\cdot 2\pi t}$

parseval

$\Rightarrow 1/4 = \frac{2\,\text{req sum} + 1/8 + \frac{1}{\pi^2}}{\pi^2}$

$\Rightarrow \pi^2/8 = 2S + 1$

$\Rightarrow \boxed{\pi^2/16 - 1/2 = S}$

3. Consider a system with $H(j\omega) = \frac{1-j\omega}{1+j\omega}$. Let the output $y(t)$ be obtained when the system is given an input $x(t) = sin(\pi t) + sin(2\pi t)$. $T=2,\ \omega_0=\pi$ Find and plot the magnitude and phase of the Fourier coefficients of $y(t)$.

(7 marks)

$$x(t) = \frac{e^{j\cdot\pi t} - e^{-j\pi t}}{2j} + \frac{e^{j\cdot 2\pi t} - e^{-j\cdot 2\pi t}}{2j}$$

$$a_{-2} = -1/2j \quad a_{-1} = -1/2j \quad a_1 = 1/2j \quad a_1 = 1/2j$$

$$H(jk\omega_0) = \frac{1-jk\omega_0}{1+jk\omega_0}$$

$$y(t) = \frac{-1}{2j}\frac{1+2j\pi}{1-2j\pi}e^{j(-2)\pi t} - \frac{1}{2j}\frac{1+j\pi}{1-j\pi}e^{j(-1)\pi t}$$

$$+ \frac{1}{2j}\frac{1-2j\pi}{1+2j\pi}e^{j(2)\pi t} + \frac{1}{2j}\frac{1-j\pi}{1+j\pi}e^{j(1)\pi t}$$

